EV Customer Prioritization

Individual project proposal | Nu Quynh Chau Nguyen | October 11 2026

I propose a decision support tool that helps an EV dealership choose which customer profiles to review first when its outreach team can contact only K people. The tool ranks profiles, displays relevant customer information, and exports a contact shortlist. A staff member makes the outreach decision.

Business problem and intended user

The stakeholder is a dealership sales or customer relationship manager with limited contact capacity. The scenario is illustrative; no partner dealership or real customer campaign is claimed. The question is: Which K profiles are most likely to have Will_Buy_EV = Yes in the approved dataset?

Charging access and daily driving requirements are practical EV purchase considerations in U.S. Department of Energy guidance [2]. The IEA also identifies home, workplace, and public charging access as relevant to EV adoption [3]. These sources support discussing charging needs; they do not establish that this dataset represents a particular dealership.

Proposed approach and intended action

Use supervised binary classification, with Logistic Regression as the main candidate, compared with a Dummy baseline and a Decision Tree. Convert model scores into an exact K-profile ranking. The Streamlit prototype lets a user upload a CSV, set contact capacity, inspect age, income, gender and driving or charging information, and download results. Consultation questions should confirm customer needs rather than assume preferences from demographics.

What success means

Precision@K measures the share of Yes labels among the K selected profiles. Lift@K divides that share by the overall Yes-label share in the evaluation set. Primary model selection uses validation Lift at a 10% contact capacity; 1%, 5%, and 20% are sensitivity checks. A useful result improves ranking over the baseline and can be reproduced by another person. No dealership conversion target or ROI has been invented.

Scope and boundaries

The approved Kaggle competition supplies synthetic profiles [1]. A score estimates its supplied label, not a confirmed purchase, extra sales caused by a call, or revenue. The data have no names or phone numbers. The current product is a class prototype; a real deployment would need permitted contact data, external evaluation, and human review.

Repository and reproduction instructions: https://github.com/Nuquynhchaunguyen/ev-customer-prioritization. Complete and attach a newly signed course validation form before submission; independent peer verification remains pending.

Analytics topics considered

The examples below are illustrative uses. Each connects a method to a decision rather than claiming a deployed system.

Topic	Realistic example	Decision supported
Supervised learning	A dealership learns from labeled customer profiles to estimate an EV purchase label.	Prioritize a fixed number of profiles for staff review.
Unsupervised learning	A retailer groups customers by shopping and service needs without outcome labels.	Design different support or outreach strategies for each group.
Dimensionality reduction	An operations team summarizes many correlated vehicle or charging measurements.	Create a simpler view while checking what information is lost.
Natural language processing	A service team categorizes written EV support requests by topic.	Route charging, maintenance, or financing questions to the right team.
Semantic or sentiment analysis	A brand team examines permitted public EV reviews for opinions about charging.	Identify concerns requiring clearer information or follow-up research.
Digital channel analytics	A dealership compares permitted campaign engagement summaries across channels.	Choose which messages or posting times to test next.

Why this project uses supervised learning

The chosen competition has an explicit Yes/No target and structured profile fields. Supervised learning directly supports a labeled-outcome ranking question. Logistic Regression supplies a simple probability score, while the Decision Tree is a comparison that can capture nonlinear relationships [7]. Unsupervised grouping could support a later descriptive study, but it does not by itself estimate this supplied purchase label.

Why the other topics are not the main method

This dataset contains no review text, support messages, campaign channel histories, or observed contact outcomes. Using NLP, sentiment, or channel analytics as the main method would require another suitable dataset. Dimensionality reduction is not necessary to start with thirteen predictors; it would add complexity before evidence shows a need.

Project focus

The focus is supervised customer prioritization for an illustrative EV dealership. The approved competition stays the project data source. The business output is a capacity-limited shortlist with context for a staff member, rather than a general algorithm tutorial or an automated customer contact campaign.

Supervised customer prioritization

From binary response modeling to an EV outreach decision

Learn from labeled examples, estimate a probability, and use the ranking to support a human decision. Logistic Regression is the project's main candidate; historical milestones below describe that method and a related industry application.



Berkson [4]
Logistic function applied to biological response modeling.

Cox [5]
Binary response regression links outcome probabilities to explanatory variables.

Moro et al [6]
Bank telemarketing research compares classifiers and evaluates targeting with lift.

This project
Synthetic EV labels support a ranked shortlist for dealership staff review.

Practical use and EV industry connection

Campaign research
A Portuguese banking study used response prediction and lift to examine who to contact [6]. This is an industry example, not evidence of EV sales impact.

EV consultation
Daily travel and home, workplace, or public charging options are practical customer needs [2,3]. The EV fields can help a staff member prepare relevant questions.

Potential value and costs

Capacity
Review a shortlist sized to the team's contact capacity.

Human judgment
Staff confirm needs, permissions, and suitability.

Resources
Setup, review, validation, and monitoring require time.

Connection to the selected dataset

The competition supplies 668,665 labeled profiles and thirteen predictors [1]. A model learns their supplied Yes/No labels; Streamlit turns its scores into a reviewable top-K list. Benefits are potential uses, not guaranteed cost savings. Synthetic labels and demographic fields require caution before any real campaign.

Figure 1. Topic and industry infographic. Timeline claims: [4-6]. EV context: [2,3]. Dataset: [1]. All icons and the diagram are original vector drawings; no outside logos are used. Full citations appear on page 8.

Dataset structure and quality

Approved source | Kaggle Playground Series Season 6 Episode 9 [1]

One row represents one synthetic customer profile. The labeled training file contains id, thirteen predictors, and Will_Buy_EV. The competition test file has the same predictors but no target; it supports scoring and demonstration, not measured evaluation.

File	Rows	Columns	Role
train.csv	668,665	15	Labeled development data
test.csv	286,571	14	Unlabeled scoring data
sample_submission.csv	286,571	2	Kaggle output template, not customer input

Key fields and definitions

Six numeric predictors: Age; Annual_Income_USD; Daily_Commute_km; Number_of_Cars_Owned; Charging_Stations_Near_Home; Charging_Stations_Near_Work. The column names indicate years, USD, kilometres, and counts where applicable, but country, income definition, and station radius are not supplied.

Seven categorical predictors: Environmental_Concern_Level, Gender, City_Type, Current_Car_Type, Home_Charging_Possible, Subsidy_Available, and Range_Anxiety_Level. Concern codes 1-5 have unverified survey anchors; the model treats them as categories. id is excluded from predictors. Target mapping is No=0 and Yes=1.

Quality checks and preparation decisions

Check	Observed finding	Decision
Missing values and blank text	0 in train and test	No imputation needed for these data
Duplicate IDs or profiles	0 in checked train and test	Preserve all rows
Checked numeric and category domains	No violations detected	Keep observed valid values
IQR flags in labeled data	Income 3,678; commute 41; cars 13,489	Retain: an IQR flag is not proof of an error
Test commute range	2 values exceed train maximum 98.7 km	Retain and warn during app scoring

Table 1. File roles and quality findings from docs/reports/tables and docs/reports/data_manifest.json. Cleaning removed zero rows and changed zero predictor values. Lossless integer storage and the reversible target mapping improve consistency without inventing observations.

Download through ev_project/download_data.py using your own Kaggle credentials and required competition access. Follow the competition rules and source license before redistributing. Keys and full raw CSVs are excluded; the repository supplies source instructions, a dictionary, and a small synthetic demo.

What the first exploration shows

Descriptive evidence from all 668,665 labeled synthetic profiles

Class balance

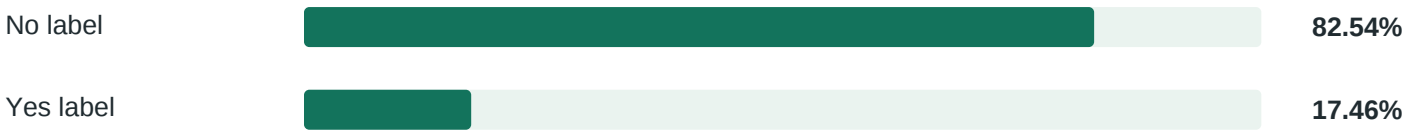


Figure 2. Yes labels account for 116,779 profiles; No labels account for 551,886. Predicting No for everyone would be about 82.54% accurate, yet it would identify no Yes profiles at a 0.5 threshold. Ranking quality is therefore more relevant than accuracy alone. Source: target_summary.csv.

Yes label share by recorded condition



Figure 3. Home charging: 12.71% of 205,988 No profiles versus 19.58% of 462,677 Yes profiles have a Yes target. Subsidy flag: 0.58% of 248,756 No profiles versus 27.47% of 419,909 Yes profiles. These associations motivate examining the fields, but changing a flag is not shown to cause a purchase. Source: categorical_target_associations.csv.

Yes label share by income group



Figure 4. The bottom income decile has 4.45% Yes labels (n=66,942); the top decile has 33.73% (n=66,754). This is an association in synthetic data, not proof that a person can afford a vehicle. Confirm budget and preferences directly. Source: income_decile_target_associations.csv.

Interpretation limits that affect the next step

No purchase timeframe or contact outcomes are supplied. Some categories are rare: High range anxiety has 2,194 profiles and only three Yes labels. Environmental concern codes must not be renamed low or high without verified definitions. Descriptive associations do not replace model evaluation or subgroup review.

Evidence the approach can work

The current prototype goes beyond a proposal; results below are internal feasibility evidence.

A fixed seed 42 stratified split assigns 468,065 labeled profiles to training, 100,300 to validation, and 100,300 to holdout. During candidate comparison, preprocessing learns only from training. Validation chooses the model. The selected model is then refitted on training plus validation and evaluated on holdout [7].

Validation candidate	Precision at K	Lift at K	ROC AUC
Dummy baseline	17.69%	1.01	0.5000
Logistic Regression	79.10%	4.53	0.9382
Decision Tree	78.76%	4.51	0.9367

Table 2. Validation n=100,300; K=10,030, or 10% of profiles. Precision@K is the proportion of Yes labels in the shortlist. Lift@K compares it with the validation Yes-label rate of 17.46%. ROC AUC describes overall ordering across thresholds. Source: docs/reports/modeling/validation_metrics.csv and validation_ranking.csv.

What the baseline comparison tells the stakeholder

The Dummy model gives every profile the same score; its fixed label-independent tie order has no meaningful feature-based ranking. Its shortlist performs near random selection. Logistic Regression ranks Yes labels more effectively on validation, but the Decision Tree is close. Selection follows the stated highest validation Lift@10% rule, not a claim that one model is universally best.

Existing holdout result and its boundaries

The already evaluated holdout top 10,030 contains 7,902 Yes labels: 78.78% Precision@K and 4.51 Lift@K. This is an observed label concentration in this synthetic evaluation set, not an expected dealership campaign conversion rate. Full-data descriptive exploration occurred before splitting, so this is an internal holdout excluded from fitting and selection, not an untouched external population.

Working artifact and proof of concept

The current project saves the fitted preprocessing and model together, reloads them, and checks that probabilities match. Streamlit loads that saved model to rank uploaded profiles. Thirty automated checks cover model separation, input validation, repeatable rankings, exact K, and correct profile attachment. These checks do not replace independent peer reproduction. Current evidence is available in notebooks, HTML reports, model metadata, and tests.

What remains unknown

Real customer performance, probability calibration outside this synthetic source, fairness of demographic predictors, and the effect of outreach remain unestablished. Further work should evaluate subgroup errors and feature choices, test the handoff on another computer, and incorporate professor feedback without repeatedly optimizing against the holdout.

From data to a staff decision

Completed prototype steps and planned extension work are labeled separately.

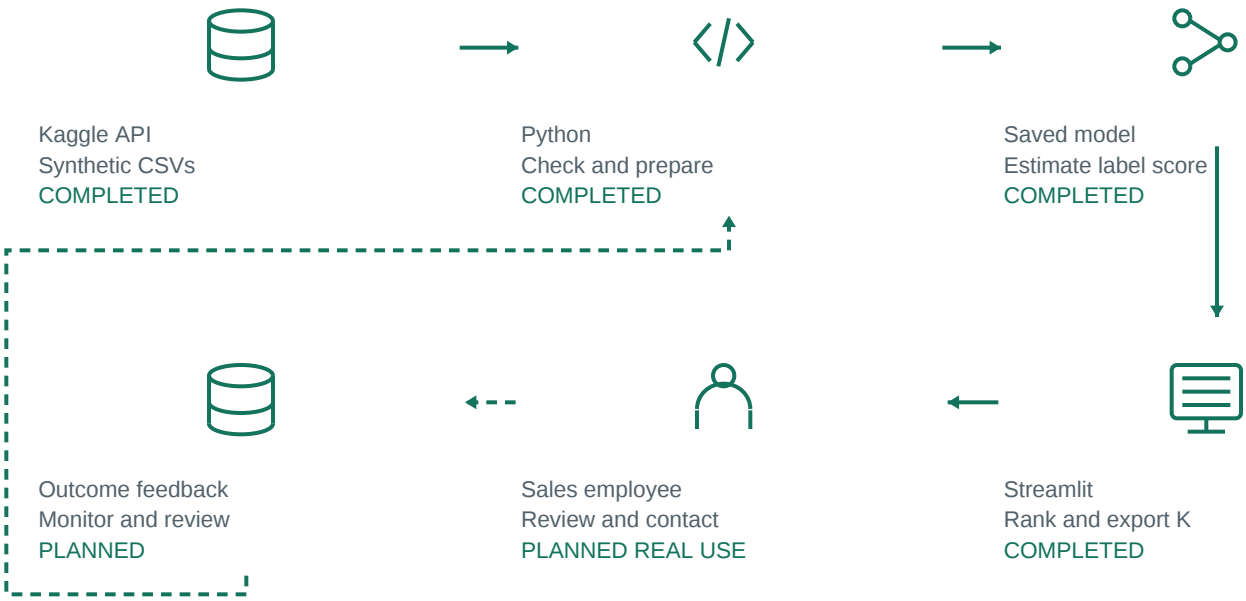


Figure 5. Complete data-to-action flow. Solid arrows show the current technical flow and proposed staff use; dashed arrows show planned outcome feedback. No actual customer contact or live sales feedback has been collected. Icons are original vector drawings.

Next work for P2 and final delivery

Incorporate P1 feedback, write the business report, update stakeholder slides, and complete peer replication. Review subgroup performance and predictor choices before any real use. Preserve the model-selection and holdout boundaries.

Plan for all three optional extensions

Extension	Planned implementation	Verification evidence
FastAPI	Scoring endpoint reuses the saved pipeline and input rules.	Valid and invalid requests tested; endpoint, example request, and run steps documented.
Docker	Versioned image contains pinned packages and fitted artifact.	Clean build and service launch; sample scores match; build and run steps documented.
MLOps	Compare input quality and drift with a saved training reference.	Shifted-input test produces aggregate alerts, versioned evidence, and investigation rules.

Table 3. All three extensions are planned, not yet implemented. Monitoring input drift is distinct from measuring prediction quality; real performance monitoring needs new verified outcomes.

Sources and reproducible access

Numbered citations support external facts; project evidence is identified by its repository path.

[1] Kaggle. Predicting Electric Vehicle Purchases. Playground Series Season 6 Episode 9. Approved synthetic competition source. Source link.

[2] U.S. Department of Energy. Alternative Fuels Data Center. Electric Vehicles for Consumers. Guidance on daily driving and charging options. Source link.

[3] International Energy Agency (2026). Global EV Outlook 2026. Electric vehicle charging. Context on home, workplace, and public charging. Source link.

[4] Berkson, J. (1944). Application of the Logistic Function to Bio-Assay. Journal of the American Statistical Association, 39(227), 357-365. Source link.

[5] Cox, D. R. (1958). The Regression Analysis of Binary Sequences. Journal of the Royal Statistical Society Series B, 20(2), 215-232. Source link.

[6] Moro, S., Cortez, P., and Rita, P. (2014). A data-driven approach to predict the success of bank telemarketing. Decision Support Systems, 62, 22-31. Author preprint and the UCI Bank Marketing record support the industry example. Source link.

[7] Scikit-learn documentation. Logistic Regression, preprocessing pipelines, DummyClassifier, DecisionTreeClassifier, and model persistence. Installed versions are pinned in requirements.txt and model_metadata.json. Source link.

[8] Solis, S. MSBA265 Final Project Instructions. Professor-provided FInal-Project.docx, received October 11 2026. P1 requirements and optional FastAPI, Docker, and MLOps rubric.

Data provenance and supporting evidence

The competition files were obtained through the Kaggle API; access instructions and attribution are in README.md. This handoff omits credentials and full raw data. The supplied demo consists of synthetic unlabeled competition profiles. Reproduce the full analysis by downloading the three CSVs, installing the pinned dependencies, and running the notebook runners. Read existing HTML reports without Python.

Evidence: docs/reports/tables/file_summary.csv, target_summary.csv, data_quality_audit.csv, iqr_outlier_audit.csv, categorical_target_associations.csv, income_decile_target_associations.csv; docs/reports/data_manifest.json; docs/reports/modeling/validation_metrics.csv and validation_ranking.csv; models/model_metadata.json. The EDA manifest describes its earlier snapshot, while model metadata records the later trained artifact.

Individual responsibility and review

This is an individual project. I am responsible for understanding and explaining the business question, data definitions, modeling choices, evidence, and recommendations. AI assistance was used to draft code and project materials. I must review the work and address instructor feedback. Independent peer verification and signatures remain pending on the separate official course form; automated tests are supporting evidence only.

Project repository: https://github.com/Nuquynhchaunguyen/ev-customer-prioritization
Official verification form: professor repository source. Web sources accessed October 11 2026.